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The evolution of human ritual behavior as a cooperative signaling platform

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ABSTRACT

Collective ritual is virtually omnipresent across past and present human cultures and is thought to play an essential role in facilitating cooperation, yet little is known about its evolution in the hominin lineage. We examine whether collective ritual could have evolved as a complex signaling system facilitating mutualistic cooperation under socio-ecological pressures in the Pleistocene. Specifically, we identify similarity, coalitional, and commitment signals as the building blocks of the contemporary signaling systems in hunter-gatherers and trace the presence of these signals in non-human primates and the hominin archaeological and paleoanthropological record. Next, we establish the underlying cognitive mechanisms facilitating these signals and review the evidence of the earliest presence of these mechanisms as well as evidence for selective pressures on the evolution of cooperative communication. The synthesis of these streams of evidence suggests that ritualized cooperative signals might have first evolved in the Early Pleistocene in the form of similarity signals, whereas coalitional and commitment signals would start appearing in the early and late Middle Pleistocene until, eventually, coalescing into a signaling system. By the arrival of *H. sapiens*, it is possible that collective ritual as a staged and repetitively performed signaling act constituted an important adaptation facilitating collective action.

KEYWORDS

African hominins; collective ritual; complex signaling systems; cooperative communication; mutualistic cooperation; neurocognitive mechanisms; pleistocene

1. Introduction

Collective ritual is one of the universals of the human behavioral repertoire, present in various social formations, be it religious groups, violent gangs, or grassroots movements. However, the omnipresence of collective ritual repeatedly puzzled evolutionary scholars because ritual participation requires substantial resource investments while seemingly not providing any apparent immediate benefits. For instance, ritual participants directly waste resources such as in animal sacrifices or threaten their lives during rites of passage where recruits undergo strenuous ordeals. Even milder forms of ritual participation require allocating time and energy that could be dedicated elsewhere. Why are people willing to voluntarily decrease their immediate well-being (as a proxy for long-term fitness) by participating in collective rituals?

Given the overwhelming spatiotemporal and cross-cultural presence of ritual behavior, an evolutionary explanation would suggest that proportional benefits must offset these fitness costs. Following a long tradition of ritual studies (Durkheim, 1912; Robertson Smith, 1898), recent

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decades brought an upsurge of functionalist explanations of collective ritual, with the most prominent explanation pertaining to ritual's ability to facilitate inter-personal cooperation (see Lang & Kundt, 2020 for a review). Using this evidence, some proposed that ritual behavior is an adaptive trait stabilizing cooperation, which likely evolved from proto-ritual behaviors in the hominin lineage (Alcorta & Sosis, 2005; Rossano, 2015; Sterelny, 2017). However, none of these accounts provide an evolutionary analysis of how ritual evolved from fixed indices characteristic of non-human animal communication into a complex signaling system including symbolic communication, thereby missing a crucial piece of evidence for the ultimate cause of ritual behavior.

The primary reason for this theoretical lacuna is the lack of archaeological evidence: inferring the evolution of behavioral traits that do not necessarily leave material traces is inescapably speculative. Yet, given the importance of ritual behavior in human lives across the world, we believe that researchers should start addressing the question of the ultimate cause of human ritual behavior using all available tools. Attempting to incept this investigation, we provide an account of the possible evolutionary pathway of ritual behavior in the hominin lineage by cross-corroborating the up-to-date available evidence from multiple disciplines.

Specifically, we first determine the primary function of collective ritual—communicating cooperative intentions via multiple signals—and review supportive evidence for this function. Next, we examine the presence of possibly homologous cooperative signals in non-human primates, finding only rudimentary signals, which suggest that collective ritual as a complex signaling system evolved only in the hominin lineage. Reviewing the archaeological record for the earliest evidence of ritualized signals suggested that the evidence is biased toward Upper Paleolithic dates, possibly obscuring the evidence of these behaviors in earlier *Homo* species. To circumvent the biased archaeological record, we focus on the evolution of neurocognitive structures that facilitate human ritualized signals and compare the estimated dating of the presence of these structures with the dating of increased selective pressures on collective action (related to variable wet-arid cycles and demographic expansion). Synthesizing this evidence, we sketch a possible timeline for the evolution of ritualized cooperative signals under various socio-ecological selective pressures and identify further areas of investigation that would advance this evolutionary model.

2. The form and function of collective ritual

2.1. The problems of living in a group

The most important benefits of living in groups among primates are the defense against predators and competing groups (Van Schaik, 1983). However, living in groups also poses challenges ranging from simple coordination difficulties, over increased resource competition, to the risk of free-riding and conflict between individual interests. In the primate taxa, where within-group interactions are relatively intense (compared to other group-living species such as bovids), these issues intensify with decreasing genetic relatedness (Hamilton, 1964) and decreasing recurrent interactions (Trivers, 1971), trends that are both characteristic of contemporary hunter-gatherers (Hill et al., 2014). In other words, increasing group size poses challenges to collective action that hinges on a reliable assortment of individuals with cooperative intentions.

An obvious candidate for the communication of cooperative intentions facilitating assortment in *H. sapiens* is language, the development of which is often championed as the defining moment of human evolution. However, as we show below, fully evolved language capacities substantially lagged behind the onset of collective action problems in early *Homo*. An assumed precursor of language—gestural communication that relies on pantomime and pointing (Arbib et al., 2008)—could address this issue, but the challenge is to explain how an intangible concept of cooperative intention could be communicated using gestures. The second challenge is to explain why this mode of communication would be preserved after the onset of symbolic

communication. We suggest that ritualization of communicated intent answers the first challenge and physical performance of the ritual signal (as opposed to the mere ethereal nature of language) the second challenge.

2.2. Collective ritual as a communication platform

We define ritual based on its formal and structural aspects that are cross-culturally less variable than its content and may, therefore, hold the key to understanding the evolutionary dynamic behind the ritual behavioral pattern. Specifically, we utilize the definition of ritual suggested by Rappaport (1999, p. 24), who considered ritual to be “the performance of more or less invariant sequences of formal acts and utterances not entirely encoded by the performers.” Rappaport recognized that ritual often does not have any practical consequences on the state of the world (i.e., it is non-instrumental) but instead offers a conventional (i.e., invariant and encoded by previous generations) and performative (i.e., visible, tangible, physical) way to communicate personal intentions toward cooperative actions stipulated by the group (i.e., the intensity of performance inter-individually vary).

Inspired by Peirce’s semiotic theory, Rappaport argued that due to the necessity of performance, ritual creates a causal relationship, that is, an index, between the material *signifier* (e.g., the number of pigs sacrificed) and the immaterial *significatum* (e.g., the extent of commitment to help in subsequent warfare). These indexical messages afford to unambiguously communicate the internal states of the performer via conventionally established channels that the receivers understand. Indeed, intense ritual performance cannot help but indicate intense support for the social order and its related cooperative stipulations, while tepid performance cannot help but indicate tepid support. Due to its reliance on physical performance that often cannot be easily faked (e.g., initiation rituals), rituals remain an important platform for reliable communication of intentions and cannot be fully substituted by symbolic communication where intentions can be lied about (Sosis, 2003). By publicly performing a ritual, group members acknowledge the existence and acceptance of social order, indicate the amount of their commitment to the order, and perpetuate the order’s key contours.

Following Rappaport’s insights, we argue that this multivalent communicative function of ritual behavior is characteristic of complex signaling systems where various signals form a functional unit (Barker et al., 2019; Hebets et al., 2016). Collective ritual fits well the criteria for a signaling system because the ritual signals are often redundant (e.g., repetitively confirming commitment to a religious order with the same utterance), degenerate (e.g., showing commitment by sacrificing money and suffering pain in the same ritual), and pluripotent (e.g., signaling both commitment and artistic creativity when piercing the body with decorated needles). However, to understand how this signaling system evolved to fulfill its central function highlighted by Rappaport (communication of cooperative intentions), we need to reverse-engineer the process of the system’s complexification. That is, we need to identify signals forming the building blocks of the signaling system and trace their origins to an exaptive pool (*sensu* Gould, 2002, p. 1271) of proto-ritual traits that evolution molded into collective ritual.

2.3. Three types of signals

We start by defining a signal as a display that is selected for its effect on the behavioral pattern of the receiver, which promotes adaptive benefits for both the signaler and receiver (paraphrased from Hebets & Papaj, 2005). For a display to be a signal, it needs to convey information (e.g., cooperative intention) that is otherwise hidden, would not be evident without such a display, and is understood by a receiver. Importantly, to be a suitable candidate for the early transitional signal, this display should function without reliance on symbolic communication. This latter condition is a crucial qualifier if we assume that ritual evolved from indexical communication analogous to the fixed signals of non-human primates (e.g., sexual swelling indicating estrus).

While contemporary rituals blend symbolic and indexical communication where signals are necessarily entangled in the dense networks of culturally specific meanings, we need to expose those signals that could serve as transitory, communicative acts between fixed indices and purely symbolic signs (Deacon, 1997, p. 72). Presumably, such signals would harness a temporal correlation between the ritualized communicative act and a subsequent behavioral pattern corresponding to a cooperative intention (Tomasello et al., 1997), allowing the receiver to form this association and select an appropriate behavioral response (Rowe, 1999). In line with this presumption, these signals would first function on the dyadic level (as in other non-human primate gestures) and only later evolve to allow for one-to-many communication of cooperative intentions.

Here we identify three types of signals that interact to facilitate cooperative communication in contemporary rituals but may have pre-dated symbolic communication: similarity signals, coalitional signals, and commitment signals (see Table 1). While similarity signals indicate shared interests between the signaler and the receiver, coalitional signals indicate membership in a specific group/collaborative ensemble, and commitment signals indicate the graded obligation toward a particular social contract. These cooperative signals move from the simplest indication of vaguely aligned interests between the signaler and the receiver (e.g., willing to engage in joint action) over a more flexible indication of partaking in a specific collective action with the receiver (willing to engage in antelope hunt) to indicating the strength of commitment to collective action with the receiver where free-riding is a threat (unequivocally will engage in antelope hunt).

Although all three signals interact as a functional unit to facilitate cooperative assortment in contemporary rituals (e.g., some commitment signals are also coalitional signals), we will treat them as ideal types throughout the article to show how they might have co-evolved from a dyadic signaler-receiver relationship to create the signaling system called collective ritual. Importantly, these ideal types refer to the original function of the signal corresponding to the then available neurocognitive abilities. As we showed briefly in Table 1 and will show more below, with further evolution of neurocognitive structures, new signals, and cultural practices, the informative potential of these signals would continually broaden (e.g., after the evolution of shared intentionality and recursion, group-specific similarity signals would indicate not only similarity but also group membership). Before providing evidence for each signal's functional role in cooperative assortment, we need to supply at least some illustrations of these signaling behaviors in the ethnographic record.

Table 1 . An overview of early ritualized signals on the dyadic signaler-receiver level and their further co-evolution with symbolic, one-to-many communication.

Similarity signals	<i>Example</i>	Movement mirroring
	<i>Signaled information</i>	Mutual relatedness with the receiver
	<i>Selective pressures</i>	Cooperation with an increasing number of non-kin (e.g., food sharing)
	<i>Facilitating mechanism</i>	Imitation scaffolded by the mirror-neuron system
	<i>Further signal co-evolution (multimodality)</i>	Mutually signaling collectivity with a group-specific synchronous action (e.g., traditional dance)
Coalitional signals	<i>Example</i>	A shortened string of gestures resembling particular action
	<i>Signaled information</i>	Intention to partake in a joint action with the receiver
	<i>Selective pressures</i>	Cooperation in a joint action (e.g., moving carcass to home base)
	<i>Facilitating mechanism</i>	Shared attention and recursion
	<i>Further signal co-evolution (multimodality)</i>	Mutually signaling membership in a joint-action coalition using a conventional symbol (e.g., spear decoration before a hunt)
Commitment signals	<i>Example</i>	Painful body modification
	<i>Signaled information</i>	Commitment to a risky joint action with the receiver
	<i>Selective pressures</i>	Cooperation in an increasingly high-stakes joint action (e.g., collective raid) and with less known others (e.g., from other groups)
	<i>Facilitating mechanism</i>	Foresight and understanding others' utility calculations
	<i>Further signal co-evolution (multimodality)</i>	Mutually signaling commitment to a group's symbol (e.g., animal sacrifice to a deity) that represents cooperative norms regulating participation in a risky joint action

2.4. *Ethnographic excursus*

Among the!Kung San, the meeting of different camps is accompanied by several days of dancing in a figure-eight pattern around the fire that “... brings the camps together to resolve their conflicts” (Katz, 1982, p. 38) as would be expected of similarity signals. Dance is the most important ceremony for the!Kung San, accompanying puberty rites or hunting (with action-specific dance indicating coalitional membership). The commitment signaling function of dance is especially pronounced during healing ceremonies where dancers heal an ill or possessed person by entering trance, which often leaves the dancers unconscious for several hours, or by setting their hair on fire and similar dangerous and energetically demanding practices (Marshall, 1962). In this manner, the healers indicate the intensity of their willingness to help another person or the whole camp if the dance is directed to generally ward off any evil.

Another example of a ritual comprising all three signals can be found in the initiation ceremony of the Andamanese people (Radcliffe-Brown, 1922). During the ceremony, the boy's body is smeared with burned iron oxide that produces a red color to indicate his membership among the initiates (coalitional signal), and his body is scarified (a mix of commitment and coalitional signals) while older men dance in synchrony (similarity signal) around the boy (Radcliffe-Brown, 1922, p. 100). Subsequently, the boy needs to abstain from certain foods for several weeks, and his body is further scarified, indicating commitment intensity.

As a third example, the Sun Dance of the Blackfoot people offered a complex arena for signaling status, chastity, as well as pro-group sentiments and commitments due to elaborate role division. Lasting for several days, the Sun Dance was a gathering place for several bands that would exchange crucial information and resources and dance in unison (similarity signal; Wissler, 1918). For example, women could take public vows of chastity in the Ceremony of the Tongues or become a medicine woman, which, while prestigious, meant giving away most of the accumulated property to others (commitment signal; Wissler, 1918, pp. 231–233). On the other hand, men could take several roles indicating their status and membership (e.g., in a war party – coalitional signal), the most severe being participation in the Torture Ceremony (commitment signal). Participants' bodies were painted to denote their role in this ceremony, pierced with wooden skewers, and later tied to a central pole of the dance lodge, where participants danced until their flesh tore apart (Wissler, 1918, p. 263).

These brief examples illustrate how each described signal fits into a general scheme of the collective ritual to inform other ritual participants about some hidden quality of the performer. While the examples were limited, a more thorough probe into the ethnographic record should identify analogous practices across the globe (e.g., using pigments to paint the body of warriors in rituals preceding a battle was traditionally found across the Americas; Laubin & Laubin, 1989).¹ Note that these signals overlap and interact in the provided examples. We should expect such interactions in a complex signaling system (e.g., synchronous dancing several hours before a raid signals aligned interests, membership in the raiding party, and commitment to the raid by the spent energy during dancing); nonetheless, we assume that their original evolutionary pathways were distinct enough to warrant their study in isolation (as described in Table 1). Likewise, the provided examples indicate that ritual participants may signal much more than their cooperative intentions (e.g., status or wealth; Power, 2017a; Xygalatas et al., 2021), suggesting that these signals may have evolved for other reasons than facilitating cooperation. We deal with these alternatives in section 5 after reviewing the evidence for rituals as signals of cooperative intentions.

2.5. *Evidence for the signals' function*

2.5.1. *Similarity signals*

We presume that early similarity signals would take the form of mirroring the receiver's movements and gestures. There is ample evidence suggesting that mimicking the behavioral patterns of

participants stripped of any symbolic meaning positively affects their perceived similarity, and the same applies to synchronous movement (for a review, see Lang et al., 2017). Since similarity positively affects the perception of mutual relatedness between the signaler and receiver, similarity signals also facilitate the perception of aligned interests, which is one of the factors affecting the structure of inter-personal ties (McPherson et al., 2001).

Indeed, mimicry facilitates prosocial behavior towards the imitator (e.g., Müller et al., 2012), and this function is pronounced in collective synchronous movements such as dancing (for a meta-analysis, see Mogan et al., 2017). Similarly, ritual activities that include exact synchrony (e.g., chanting) were shown to increase monetary contributions to a shared pool in an economic game more than rituals without synchrony (Fischer et al., 2013). Ritualized synchronous activity also leads to higher ingroup affiliation among children (4–11 years old) compared to unstructured group activity (Wen et al., 2015).

Importantly, as predicted by gradual signal complexification, these similarity signals currently interact with coalitional membership. For instance, participants tend to imitate more the behaviors of members of their ingroup than their outgroup (Yabar et al., 2006). When ostracized from their group, participants tend to increase their imitative behaviors to fit back into their group (Watson-Jones et al., 2016). Likewise, infants spend more time (indicating surprise) looking at people who affiliate after performing intentionally different arbitrary actions than looking at affiliating people who performed the same action (Lieberman et al., 2018).

2.5.2. Coalitional signals

We presume that early coalitional signals would take the form of ritualized gestures resembling a specific cooperative activity and or the form of visual markings. The ritualized signal of intention to participate in a joint action would communicate to the receiver that the signaler proposes to share a specific common goal including the necessary coordinative background to reach that goal (Tomasello, 2008). Ritual behavior and ritual markings are effective in this regard because they protect the coalitional signal from ambiguity by enforcing binary distinction on otherwise continuous processes (a person either marks or does not mark their body to indicate membership in a raiding party).

Ritually established relationships appear to be the best predictor of cooperative relationships among the Ache and Hadza hunter-gatherers (Hill et al., 2014). Moreover, studies with children showed that when two groups of preschoolers were taught a different ritual (group-specific play with beads), they spent more time displaying the ritualized sequence of gestures to the ingroup members compared to children in the control condition who played with the beads in their idiosyncratic way (Wen et al., 2020).

In line with signal complexification, having the abstract notion of group identity would imply a shared coordination ground of multiple group members, and mutually signaling coalitional membership would facilitate quick and precise execution of larger collective tasks (Sterelny, 2012, p. 142). Indeed, using visual reminders of ritual participation increases the rate of cooperative exchange among anonymous co-religionists in economic games (Shaver et al., 2018), presumably by implying a shared coordinative and cooperative ground.

2.5.3. Commitment signals

We presume that early commitment signals would take the form of self-inflicted pain and resource waste before a dyadic risky joint action.² Since ritualized signals are often excessively costly, they indicate a graded assurance of commitment to a joint action that is more reliable than abstract symbolic communication. This principle is formalized in the costly signaling theory (Grafen, 1990; Zahavi & Zahavi, 1999), which holds that in situations where both the signaler and receiver have conflicting interests but would benefit from an honest communication of unobservable traits, such a signal would carry increasing production costs—relative to benefits—to secure its honesty. Since uncommitted signalers are not willing to produce the highly costly signal because they

perceive the cost as too high or benefits as too low (Lang et al., 2022), paying the cost of the signal and later garnering the associated cooperative benefits is perceived as profitable only for the committed individuals (Sosis, 2003).³

Regarding empirical support for commitment signals, paying costs to signal cooperative intentions is associated with larger contributions to the common pool in the public goods game (Lang et al., 2022) and costly rituals are recognized as commitment signals by the receivers (Hall et al., 2015). Moreover, toddlers are sensitive to the time and effort costs of others' actions and can translate these costs into the moral domain (Jara-Ettinger et al., 2015), suggesting that costly action in a cooperative context would be understood by both signalers and receivers as more binding from an early age.

In line with the assumed signal complexification, costly rituals often appear in religious context where they indicate commitment to supernatural deities that regulate social conduct (Chvaja et al., [forthcoming](#)). Costly religious acts were associated with being named as helping others in rural South India (Power, 2017b), and taking part in the extreme ritual of Thaipooosam Kavadi in Mauritius positively predicted subsequent anonymous charitable donations to the local Hindu temple (Xygalatas et al., 2013).

2.5.4. Summary of signal types

We outlined similarity, coalitional, and commitment signals as ideal types of signals that comprise the signaling system of human collective ritual. We provided evidence that these signals are associated with cooperative intentions, are understood as such by receivers, and are acquired relatively early during development. It follows that if the similarity, coalitional, and commitment signals were ritualized at the dyadic level and gradually coalesced to form a stable trait of one-to-many cooperative communication we call collective ritual, each of those signals had to be under directional selection corresponding to particular selection pressures (as outlined in [Table 1](#)).

3. Identifying the window for the evolution of rituals

To investigate the hypothesis that ritualized signals assorting cooperators evolved under directional selection, a temporal window for the evolution of these signals needs to be identified. We compare the cooperative signals possibly present in the last common ancestor of *H. sapiens* and chimpanzees with the dating of these signals in the archaeological record. This comparison will estimate when those signals might have originated in the hominin lineage. Having estimates of the earliest possible presence of collective ritual should serve as a steppingstone to looking into the preceding history of the genus *Homo* in search of the selective pressures and processes in the making of collective ritual.

3.1. Cooperative signals in non-human primates

The primate order displays a rich array of communicative signals, predominantly acoustic. However, those signals are mostly innate and inflexible displays that lack recursion, fitting into the category of fixed indices (Tomasello, 2008). While some vocalizations may be invented and later ritualized, the less-frequent gestural communication is more flexible (albeit often paired with acoustic signals in apes). As in the assumed origin of ritualized signals in hominins explained in section 2.3, invented gestures originate from co-opting behavioral patterns from one context (e.g., foraging, eating) to communicate different information (e.g., initiate grooming). Through the process of ontogenetic ritualization (Tomasello et al., 1997), a dyad shortens its idiosyncratic string of gestures to a simple gesture that later functions as a placeholder for the original complicated pattern (e.g., play initiation), eventually yielding group-specific communication convention (Fröhlich & Van Schaik, 2020; Perry et al., 2003). Theoretically, such invented gestural patterns could be utilized for cooperative communication, as outlined in the current paper.

3.1.1. Similarity signals in non-human primates

The most fitting example of similarity signals is chimpanzee pant-hoot chorusing, which represents an intentional vocal signal that co-occurs with other affiliative behaviors, is performed by socially preferred partners, and may be group-specific (Fedurek et al., 2013). Apart from this one instance of vocal synchrony, the evidence for using imitative behavior as signals in non-human primates is lacking, although there is suggestive evidence for the capacity to use imitation as affiliative signals. For example, capuchin monkeys were shown to prefer human experimenters who mimicked them as opposed to experimenters who did not (Paukner et al., 2009).

3.1.2. Coalitional signals in non-human primates

Generally, the recruitment of coalition partners in coalitional primate species relies on innate instead of invented signals, as well as on social grooming and previous coalitional experience (Fröhlich & Van Schaik, 2020; Watts, 2002). There are hints of socially learned signals among chimpanzees, such as the high-arm grooming; however, this behavior indicates rather a subordination of the arm-holder than a coalitional membership because the variation in grooming styles is traced to matrilineal rather than specific groups (McGrew, 2017). Similarly, white-faced capuchin monkeys form social traditions related to group-specific modes of dyadic interaction, but these group-specific modes are not used to discriminate between less-known interactants from other groups (Perry et al., 2003).

3.1.3. Commitment signals in non-human primates

There is relatively rich evidence for commitment-like signals in non-human primates. For instance, male Guinea baboons and macaques were observed to use genital fondling as a risky re-assurance of the stability of their dyadic bond, and the intensity of this practice was positively correlated with the social distance between the signaler and the receiver (Dal PESCO & Fischer, 2018). Among chimpanzees, behaviors such as testicle handling were recorded as re-assurance gestures (Bygott, 1974). While these behaviors do not represent a direct fitness cost, the behaviors carry significant risks by targeting specific body parts that are extremely sensitive. That is, making oneself vulnerable to the attack of others serves as an honest signal of affiliative intentions (Bruner et al., 2017). Likewise, these signals are used to enforce an affiliative signal from others — accepting the aversive stimuli (finger in mouth, testicle handling) is an index of positive relationship (Perry et al., 2003).

3.1.4. Summary of non-human primate research

While we identified several signals in non-human primates that structurally correspond to the three human signals, the non-human primate signals are rather rudimentary and play only a minor role in facilitating mutualistic cooperation. Notably, the identified signals were gestural (except for vocal chorusing), dyadic, and while sometimes costly, this cost was not graded and only tentative. Since the signals were detected across various primate taxa, it is possible that these early signals were in place in the last common ancestor of the simian infraorder and, therefore, presented a pool of signals that have been further molded into ritualized cooperative signals in the hominin lineage (where selective pressures on cooperative communication were hominin-specific; e.g., chimpanzees engage mostly in low-stake collective action). However, while the reviewed evidence is supportive of this view, a reconstruction of the phylogenetic tree of these signals would be needed to establish whether the signals represent a synapomorphy or are the product of convergent evolution in various species of the simian infraorder.

3.2. Archaeological evidence for cooperative signals

3.2.1. Similarity signals in the archaeological record

We argued that similarity signals such as movement mirroring or synchronous dance serve to enhance the perception of aligned interests. However, since gestural similarity signals would

leave no archaeological record, we are left with depictions of such activities through cave art. The oldest recorded dance scene has been uncovered in Borneo caves and dates back to 20 ka (Aubert et al., 2018). Nevertheless, the neurocognitive machinery necessary to produce cave art is far superior to the machinery assumed for similarity signals, thus substantially biasing these estimates to younger dates.

3.2.2. Coalitional signals in the archaeological record

The oldest material remains of coalitional signals are related to the use of pigments since these signals (as opposed to gestural signals) leave material records. Although pigment use, such as red ochre, was documented across hunter-gatherer populations in various contexts, coalitional signaling appears to be the most important function (Duarte, 2014). Pigment use was recorded as early as 285 ka in Kenya (Deino & McBrearty, 2002). There are also indications of pigment used in the Fauersmith industry from South Africa, dating back to 500 ka (Watts et al., 2016).

3.2.3. Commitment signals in the archaeological record

Costly commitment signals might be found in intentional body modifications; however, these modifications are ordinarily performed on the soft tissue, thus leaving us again only with artistic depictions of these practices that date back to 30 ka (e.g., the pierced phallic statuette from Vogelherd; Angulo et al., 2011). Similarly, hand images found across Spain and France printed on cave walls (dating back to 27 ka) that miss various finger segments may indicate costly commitment signals (McCauley et al., 2018).

3.2.4. Summary of the archaeological record

The oldest archaeological evidence for similarity and commitment signals dated only to the rich symbolic behavior of the Upper Paleolithic and mainly was found outside Africa. We conclude that the young material evidence testifies more about our ability to document the use of such signals than about the ability of past hominins to produce them. Indeed, the oldest material evidence for ritualized signals coming from Africa is pigment use (500–285 ka), which does not rely solely on behavioral gestures or perishable material such as human soft tissue.

4. The evolution of collective ritual in the hominin lineage

To further examine the evolutionary pathways of cooperative communication, we now turn to the evidence indicating the presence of the key neurocognitive mechanisms necessary for sending and decoding the three types of ritualized signals. This dating may provide a rough outline of when each ritualized cooperative signal would be feasible in the hominin lineage (possibly circumventing the bias in the archaeological record). To refine these estimates, we next investigate the available evidence for the primary selective pressures on collective action in the hominin lineage that would force to co-opt these neurocognitive mechanisms to serve the signaling function.

Note that we focus solely on the evidence from the African continent in this review. Although some of the examined evidence has earlier dates outside Africa, we remain conservative in not assuming that this evidence indicates the earlier presence in Africa. Many examined traits would be crucially dependent on specific ecologies and subject to convergent evolution. Thus, to establish the pathways of traits preceding collective ritual in *H. sapiens*, we trace the presence of these traits in African hominins (rather than assuming that these traits were brought to Africa by hominins evolving outside of that continent).

4.1. The evolution of mechanisms facilitating cooperative signals

4.1.1. Mechanisms underlying similarity signals

Similarity signals crucially rely on the ability to imitate others where the imitator needs to transpose the movements of the imitatee onto their motor network (the correspondence problem). This function is supplemented by the mirror-neuron system (Gallese et al., 1996; Heyes & Catmur, 2020), which translates the audio-visual inputs generated by the action of others into motor commands and maps them onto the imitator's pre-motor network. Crucially, imitation patterns may be modulated top-down by the medial prefrontal cortex, indicating that these patterns can be used as an intentional signal (Wang & Hamilton, 2012). The ability to imitate others is further strengthened by motor learning (Heyes, 2011) and mediated by greater mirror-neuron system activity, as seen in professional performers observing their own movements (Calvo-Merino et al., 2006). A frequent inter-personal imitation would therefore promote a recognition of aligned interests based on enhanced perception of similarity (Krueger, 2013).

Importantly, high-fidelity imitation (more specifically overimitation) is crucial for social learning and is the cornerstone of human cumulative culture, largely absent in other non-human primate species (Tennie et al., 2009). We can infer on the presence of high-fidelity imitation in the hominin lineage by examining the onset of imitative social learning, which appears to be first necessary for transmitting the toolmaking process of the Acheulean Industrial Complex (Shipton & Nielsen, 2015) that dates back to 1.76 Ma in West Turkana, Kenya (Lepre et al., 2011). The Acheulean industry involves causally opaque actions that are not intuitively related to the goal of the action and require high-fidelity action imitation (Schillinger et al., 2015). Thus, with the onset of the Acheulean industry and its spread across the African continent (earliest evidence from South Africa by 1.6 Ma; Chazan et al., 2008), the presence of high-fidelity imitation might be assumed in *H. ergaster*. The co-option of the structures underlying overimitation for the purpose of signaling is further supported by the cerebral innovations found in *H. ergaster*, especially in the regions that are used for communication in contemporary *H. sapiens* (Ponce de León et al., 2021).

4.1.2. Mechanisms underlying coalitional signals

Cognitive mechanisms facilitating coalitional signaling assume dyad-specific vocal or gestural patterns at a minimum. Therefore, associative learning is a crucial ability for receivers to connect the observed signal with the ensuing signaler's action. On the signaler part, coalitional signals necessitate shared intentions and, by extension, recursion. Shared intentions are crucial for joint action because they allow actors to monitor each other's performance and dynamically adjust their actions to the actions of the other actor. Recursion describes the ability of individual A to understand that she shares a communication code with an individual B and that individual B has the same knowledge.⁴ Shared intentions and recursion crucially rely on functional neural structures traditionally identified as the theory of mind (Frith & Frith, 2006). While the basis of action understanding likely depends on the exact mechanism translating audio-visual inputs of the action of others (i.e., the mirror-neuron system), the identified neural structures employed by the theory of mind to facilitate shared intentions and recursion are generally broader (Schurz et al., 2014).

Importantly, while dyadic and triadic coalitions are observed in non-human primates, they are not cooperative coalitions in the meaning that they would spring from shared intentions and cooperative participation in goal attainment. For instance, collective chimpanzee efforts such as red colobus hunts are driven by "impact" individuals initiating the action without necessarily co-representing other actors' roles (Gilby et al., 2015; c.f. Boesch, 2002). Moreover, chimpanzees do not readily share the spoil of such hunts, and catch re-distribution relies on tolerated theft (Wood & Gilby, 2017). Compared to chimpanzee coalitions, human coalitions recruited for collective actions are highly cooperative and involve multiple roles, co-representing others and their roles and understanding shared goals (Sterelny, 2012, p. 76).

In the hominin lineage, large-game hunting is, apart from warfare, one of the most important collective actions requiring coalitional recruitment and may serve as an indicator of the presence of shared intentions. A proxy for coordinated hunting (or other aggressive collective action) is a toolkit dated to 500 ka that comprised hafted stone points, which likely functioned as spear tips for hand-held weapons in Kathu Plain, South Africa (Wilkins et al., 2012). Notably, this indirect proxy for collective action coincides with the first appearance of pigment use in South Africa discussed in section 3.2.2. However, hunting tools are more widely detected in the African record starting from 300 ka (Sahle et al., 2013), which would be a more conservative estimate.

Furthermore, one of the selective pressures for recursion would be teaching or teacher-apprentice relationship. That is, the ability to intentionally instruct another person on their activity (Sterelny, 2012). Although teaching might have been already necessary for the transmission of the Acheulean industry, it is more widely assumed to be in place in the Middle Pleistocene due to an increasing diversification of tool technology associated with new core reduction strategies to produce blades (dated to 550 ka in East Africa; Johnson & McBrearty, 2010). The subsequent transition to the Levallois industry (which appeared periodically in geographically disparate regions until widespread in Africa after 300 ka) was very likely transmitted using teaching (Lycett et al., 2016).

4.1.3. Mechanisms underlying commitment signals

The underlying neurocognitive mechanisms supporting sending commitment signals overlap to a large degree with coalitional signals. However, commitment signals are more demanding to decode: apart from understanding other people's affordances (needed for coordination during joint action), signal receivers need to understand the costliness of the performed act and weigh it against alternative behavioral choices that the signaler could have made (Bulbulia, 2009). That is, to understand that other people have a set of behavioral choices from which they opted for the particular signal intensity (Jara-Ettinger et al., 2015). Such imaginative capacity would be crucially dependent on mental time travel capabilities needed for planning future actions. Importantly, experimental evidence shows that, in contrast to 4-year-old children, chimpanzees fail to secure a reward from two alternative sources, suggesting that chimpanzees do not internally simulate alternative future outcomes (Suddendorf et al., 2017).

One source of evidence for hominin mental time travel relates to the transportation of materials to distant foraging sites that indicates foreseeing the need to make tools in the future. While the transport of tools recorded in the Oldowan industry is confined to < 10 km (Plummer, 2004; similar to chimpanzees), it is only in the late Middle Pleistocene, around 300 ka, when long-distance transport (50 km) of obsidian as a raw material for knapping is documented in Kenya (Brooks et al., 2018).

With the advance of mental time travel capabilities, similarity, coalitional, and commitment signals could extend from "here-and-now" situations where a signal is associated with an immediately following behavioral pattern to more distant future actions (e.g., cutting forearm as a sign of unequivocal support during tomorrow's raid). Together with the ability to transpose intentions from one domain (signaling behavior) to a different and temporally distant domain (future collective action), mental time travel capabilities would afford staged and regularly performed signaling events associating ritualized signals with future collective action, which is to say, collective ritual.

4.2. The evolution of ritual under selective pressures on collective action

We identified the periods when the key neurocognitive mechanisms that facilitate each ritualized cooperative signal might have been first present. However, if the ritualized signals evolved because they adaptively promoted mutualistic cooperation between the sender and receiver, we need to further show that there were selective pressures for cooperation in the hominin lineage that would push using the original mechanisms in signal production. We surmise that these pressures could have been constituted by both extrinsic factors such as environmental variability and the

associated faunal turnover (Potts, 2012) as well as intrinsic factors related to morphological, behavioral, and, later, cultural adaptations pressing on further innovations through feedback loops (e.g., better hunters live in larger societies that pressure further evolution of social cognition, which may drive technological innovation in hunting, etc.). Against the backdrop of this complex causal web, we will discuss the onset of primary selective pressures that may have triggered the evolution of cooperative communication and, subsequently, collective ritual without exploring the feedback loops between those pressures (which would constitute another study).

4.2.1. Evolution of similarity signals

To start, we dated the possible presence of early imitative skills that would afford similarity signals back to ~1.76 Ma, corresponding to the emergence of the Acheulean industry associated with *H. ergaster* in Africa (section 4.1.1). This dating coincides with the prolonged variability of moist and aridity cycles, which created substantial selective pressures on the adaptability of *H. ergaster* to various environments as well as on the ability to extract more animal tissue as a reliable source of calories (Patterson et al., 2019). Likewise, converging evidence from the comparison of cranial size, type of habitat, and feeding patterns of *H. ergaster* with non-human primates and contemporary hunter-gatherers suggest that these hominins lived in a meta-group network of around 100 members with individual foraging bands comprising 15–28 individuals (Domínguez-Rodrigo et al., 2019), which would constitute a novel selective pressure on cooperative communication (compared to a smaller social network of *H. habilis*).

Since imitation was likely in place to copy the Acheulean tool-making process, similarity signals such as movement mirroring or vocal chorusing might have been exapted from these imitative abilities (i.e., ability to precisely imitate another's motor patterns) to imply aligned interests during an increasingly more frequent interactions with less-related individuals. The inferred aligned interests would ease inter-personal cooperation and sharing of cooperative spoils.

4.2.2. Evolution of coalitional signals

The early forms of the neurocognitive structures supporting coalitional signaling, as indicated by coordinated collective action and teaching, were dated to 550 ka (section 4.1.2). Some evidence of earlier coordinated actions outside of Africa as well as estimates of cranial capacity associated with recursion (Gamble et al., 2014) suggest that these structures might have evolved already during a prolonged wet-arid variability cycle between 1.1–0.9 Ma (Potts & Faith, 2015). Nonetheless, we remain conservative and rely on the evidence for teaching since teaching would also pressure the evolution of ostensive communication in which the teacher needs to assure learners' attention (Csibra & Gergely, 2011). That is, we conjecture that it is more likely that coalitional signals would be evolving in later hominin such as the African and Eurasian assemblage of fossils labeled as *H. heidelbergensis* (Stringer, 2016).

One of the primary selective pressure on cooperative assortment in the African *H. heidelbergensis* would be an increased reliance on collectively acquired meat (Galway-Witham et al., 2019). Large-game hunting would constitute inter-dependence of group members who needed each other to acquire more meat, especially if providing for individuals with other important roles such as childcare or basecamp defense (Tomasello et al., 2012). Coalitional signals might have played a pivotal role by effectively grouping cooperative individuals for joint hunts or moving and processing animal carcasses. However, given the geographical spread of *H. heidelbergensis*, it should be assumed that this repertoire would periodically appear and disappear together with the variable selective pressures in given environments.

The signaling repertoire could have been gradually stabilized by an evolving suite of cultural technologies related to the establishment of home bases (Rolland, 2004), where ritualized cooperative signaling could be gradually woven into the fabric of everyday social life, allowing for its coalescence into a complex signaling system. In this respect, a key cultural technology would be fire use, as documented in the contemporary ethnographic record (Wiessner, 2014). Fire would

afford communal night events that might have included repetitive signaling toward multiple individuals. Regular staged performance of ritualized dyadic or triadic signals around bonfires would trigger conformist social learning (copy behavior of the majority in a group; Henrich & Boyd, 1998) and allow for the origins of group-specific signaling traditions, that is, signals understood by the whole group (affording one-to-many communication).

Of course, such collective performance need not be limited to cooperative communication. Communal fires would enable the conventionalization of other activities such as stone-tool making, cooking, or food processing, where one-to-many displays of complex procedures would allow for their group-level standardization (relying on overimitation), effectively establishing traditions with a normative appeal as well as social identity. However, in contrast to the conventionalization of these mundane activities, the staged and repetitive performance of ritualized cooperative signals (standardizing the signals into “one proper form”) would institute collective ritual as the public platform for communicating cooperative intentions. Whereas fire use dates deep to the Lower Pleistocene (Gowlett & Wrangham, 2013), the earliest evidence for the habitual use of fire associated with home base, which is key for our argument, comes from the Qesem cave in Israel at 400 ka (possibly associated with *H. heidelbergensis*; Barkai et al., 2017). By 350 ka, fire use was widespread in Africa (Macdonald et al., 2021).

4.2.3. Evolution of commitment signals

The presence of neurocognitive structures supporting successful decoding of commitment signaling was estimated around 300 ka (section 4.1.3), in a period of the graded transition from *H. heidelbergensis* to the archaic *H. sapiens* (Hublin et al., 2017). In the same period, megagrazers in East Africa were replaced by smaller-bodied herbivores, and the frequent wet-dry cycles were associated with resource patchiness and variable resource landscapes even at a scale of 20 km (Potts et al., 2020). Resource unpredictability characteristic of this climate variability would pressure the evolution of foresight to simulate possible future events and select the most secure options. Together with regular signaling performance around a communal fire that would create a robust associative relationship between the signal and future action (e.g., iconic depiction of a hunt and the next-day hunt), commitment signals might have originated as a response to doubts produced by internal simulation of others’ possible actions (especially in the case of less-known others). Co-opting the reliability-securing function of costly communication from other domains such as meat-sharing could have provided the necessary material anchoring for these newly formed signals.

The newly evolved commitment signals would be essential in between-group contact (as is the case with contemporary hunter-gatherers; Hill et al., 2014) where interactions with less-known others would be imbued with free-riding possibilities and the stakes would be much higher (e.g., battles between clans). Since cumulative cultural evolution leads to flexible adaptations to local environments, the variability of ecological zones characteristic of East Africa from 300 ka likely corresponded to variable cultural (non-kin) groups that would interact in a relatively constrained geographical area (Malinsky-Buller & Hovers, 2019). This between-group contact would gain importance with resource scarcity indicated by the sudden decrease in the effective population size of African humans post 300 ka (Bergström et al., 2020). Likewise, this scarcity would increase competition for resources with other groups, implying the increased likelihood of inter-group conflict, where reliable coalitions with other groups (secured by commitment signals) would provide an adaptive advantage. Indeed, there is evidence from contemporary hunter-gatherers showing that inter-tribal warfare is positively associated with the intensity of commitment signals (Sosis et al., 2007).

4.2.4. Summary of signal evolution

By synthesizing various streams of evidence, we arrived at a possible scenario of the evolution of collective rituals. In this scenario, collective ritual would evolve as a coalescence of three ritualized cooperative signals that communicate shared interests, willingness to partake in a joint

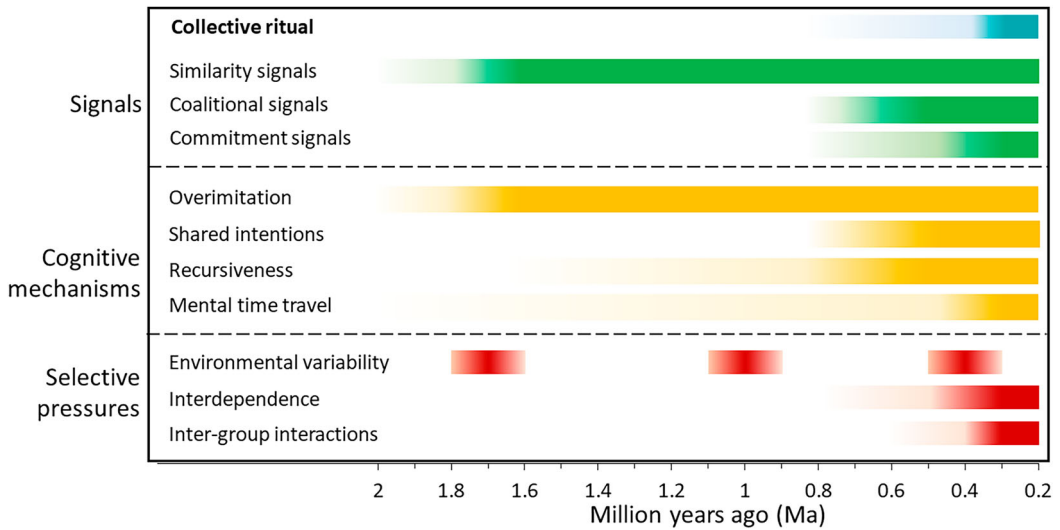


Figure 1 . An overview of the estimated onset of fundamental elements for the evolution of collective ritual in the hominin lineage. Shaded areas represent uncertainty about the presence/absence of a signal, mechanism, or selective pressure. The shading has three intervals (possible emergence, likely emergence, presumed presence) based on the timing estimates provided in the main text. The presence of signals is extrapolated from the synthesized evidence of the presence of underlying cognitive mechanisms and selective pressures. This model conjectures that collective ritual as a trait might have emerged around 300 ka as coalescence and regular performance of the three ritualized signals. However, despite the continuous shading, note that mechanisms/signals may appear intermittently and even disappear.

action, and commitment to a risky joint action (see Figure 1). Collective ritual would further co-evolve with symbolic communication such that conventional norms around resource sharing, mating, exchange, fighting, teaching, and stone-tool production would regularly co-occur with signals of commitment to those norms (as Rappaport would have it). Eventually, ritual performance would become a norm or expectation itself (e.g., initiation rituals), which would enforce revealing cooperative intentions before risky collective action (and also causally affect commitment; see section 5.1). Ritual’s function would be further enhanced by taboos, permanent markers (e.g., scarification), and similar ritualized practices associated with collective action. The lingering question of when collective ritual became the essential building block of religious systems supporting belief in supernatural agents remains open for future work, although there already are several significant proposals compatible with our proposition (Alcorta & Sosis, 2005; Rossano, 2010; Sterelny, 2017).

5. Alternative explanations for the evolution of collective rituals

We proposed that collective ritual evolved as an adaptation to selective pressures for cooperative communication, which enhanced the cooperative abilities of Pleistocene hominins. By synthesizing evidence from multiple disciplines, we aimed to illustrate the confidence or lack thereof in such an evolutionary scenario. In this section, we briefly discuss alternative scenarios for the evolution of collective ritual.

5.1. Ritual as a facilitator of social cohesion

By focusing on the communicative aspect of collective rituals, we neglected that by participating in collective rituals, people not only communicate their similarity, coalitional membership, and commitment to others but also effectively establish these qualities in themselves. Ever since Durkheim

(1912), the effervescence of collective rituals (Bulbulia et al., 2013) as well as shared dysphoric experiences of extreme ritual practices (Whitehouse, 2018) were thought to cement inter-individual bonds. Thus, the evolutionary roots of collective ritual may lie in its ability to establish bonds rather than communicate them (Rossano, 2010). While we concur that the social-bonding consequences of ritual participation are noteworthy effects, we maintain that these effects are partially derived from (a) the signaling function of collective ritual and (b) the blending of collective ritual with other bonding technologies that need not be ritualized.

First, while ritual performance informs receivers about the performer's qualities and internal states, the performance also informs the performer. Indeed, the fact that the performer is one of the crucial receivers of the ritual message was highlighted by Rappaport (1999) and elaborated in the model of religious socialization through ritual (Sosis, 2003). By partaking in the ritual, participants inform themselves about their beliefs, status, and commitments, effectively influencing their future cooperative behavior.

Second, establishing mutual ties via social grooming or play does not necessarily involve invented communicative acts because these activities trigger the endogenous opioid system that facilitates social bonding (Dunbar, 2012). The defining aspects of collective ritual, that is, exaggerated and repetitive performance of stereotyped gestures and vocalizations, directly point to the primacy of the communicative function of rituals (i.e., these aspects indicate a need for signal clarity). This conjecture is further supported by the non-human primate ritualized behaviors where ritual greetings test the existing mutual bonds rather than establish them (Perry & Smolla, 2020).

Nonetheless, once collective ritual was established as a repetitive, conventional performance, the bonding activities might have been blended with the communicative function, leading to collective performance such as synchronous dancing or chorusing that affect both perceived similarity and bond strength (Lang et al., 2017; Weinstein et al., 2015).

5.2. The communicative pluripotency of collective rituals

While we favored the cooperative communication function of collective rituals in the main paper, it is important to note that many different messages might be communicated because collective rituals are complex signaling systems (Barker et al., 2019). Apart from similarity, coalitional, and commitment signals, ritual performance may signal material wealth and social status (Power, 2017a; Power & Ready, 2018; Xygalatas et al., 2021) or even physical attractiveness (Van Slyke, 2017). While we do not doubt that various messages are communicated through the ritual platform, we maintain that the cooperative intention is the likely primary message communicated in collective rituals (see Sosis et al., 2007 for a test between these hypotheses on ethnographic data). Indeed, some sexual selection accounts assume that the communicated mate quality relates to the commitment to group norms (Van Slyke, 2017), and there is preliminary supportive evidence (Xygalatas et al., 2022). Likewise, while synchronous movement appears also in the mate-selection context, it is often a by-product of conventional ways of parading physical prowess and attractiveness of individual performers rather than an intentional signal.

5.3. Collective ritual as an extension of individual ritual

Ritual behavior may have first evolved as an individual idiosyncratic habit and only later expanded to a collective practice. A potential candidate function for the primacy of the individual-level account is the ritual's anxiolytic effect (Karl & Fischer, 2018; Lang et al., 2015, 2020; Sosis & Handwerker, 2011). The ritual decrease of anxiety might have evolved to dampen the overly sensitive anxiety subsystem (especially in the domain of uncontrollable events) by molding the product of blindly copying causally opaque actions into an exapted trait (Lang & Chvaja, 2023). Since complex cultural practices are often imitated without understanding the mechanistic function of individual sub-steps, similarly complex action sequences might have been used to influence uncertain and

uncontrollable future events (Fessler, 2006). Indeed, contemporary hunter-gatherers' pre-hunt or pre-battle rituals are permeated with taboos and superstitions concerning the correct ritual action securing success⁵, which may decrease anxiety and improve performance (Damisch et al., 2010). However, this account does not explain why collective rituals would be staged, public, and often very costly performances and would need to assume relative safety of ancestral environments (such that ritualized anxiety decrease would be adaptive) or posit a by-product scenario (Boyer & Liénard, 2006).

5.4. *Collective rituals as a spandrel*

Following previous clarifications of the adaptationist program (Sosis, 2009), we defined collective ritual, specified its function, and reconstructed its possible evolutionary history, including the analysis of the exaptive pool of mechanisms and signals that may have been co-opted under the specified selective pressures. This is not to say that all aspects of collective rituals and ceremonies are adaptive, functional, or even affect performers. Indeed, the cumulative cultural processes would ornament the crucial function with non-functional layers.

Nevertheless, we maintain that the presented evidence suggest that the core aspects of collective ritual present a complex adaptation and that it is unlikely such a costly behavior would be maintained across present and past societies if non-functional. Having said that, we are open to a re-analysis of the presented evolutionary model following Gould's conditions for identifying spandrels (Gould, 2002, pp. 1256–1257), which would show that (a) collective ritual reliably appears as a spandrel of specific psychological mechanism across cultures, (b) variation in ritual occurrence is tightly correlated with the variation of the underlying psychological mechanism, (c) collective ritual has the same form but does not gravitate to the same context (e.g., cooperative action problem) across cultures (i.e., is randomly distributed across contexts), and (d) that ritual is suboptimally designed to facilitate mutualistic cooperation.

6. Conclusion

This article opened with the bewildering omnipresence of ritual behavior, paraphrasing Rappaport's (1999, p. 140) puzzlement over why humans "who can communicate with ease, efficiency, and subtlety through language should also employ such an awkward, limited and expensive mode of communication as physical display." We attempted to answer this puzzle by proposing that ritual evolved from pre-symbolic, gestural communication of cooperative intent, and remained conserved in the hominin lineage because ritual performance facilitates reliable cooperative communication, which the ephemeral properties of symbolic communication could not supplant.

While the evidence for this evolutionary model is sparse due to the biases of the archaeological record, we hazarded a reconstruction of the evolutionary pathways of ritual behavior. Identifying the likely onset of the key neurocognitive structures that underlie ritualized cooperative signals and determining the primary selective pressures that would force adopting these neurocognitive structures for cooperative communication allowed us to estimate the possible evolutionary pathways of ritualized signals. Instead of choosing one selective pressure that would push the evolution of collective ritual, our model differs from previous accounts (Rossano, 2009; Sosis & Alcorta, 2003; Sterelny, 2017; Turner et al., 2017) because we carved the evolutionary history of collective ritual out from early dyadic signals affected by multiple selective pressures. Only later would these signals interact in feedback loops with advancing cognitive and social-organization mechanisms, leading to the complex signaling system of collective ritual we observe today.

Admittedly, while based on up-to-date, pertinent evidence⁶, the nature of this reconstruction is inherently speculative. As illustrated in Figure 1, there is ample uncertainty around the dating of the onset of individual selective pressures, mechanisms, and signals. Our main goal was to stimulate new research questions to enrich our understanding of collective ritual and its

ultimate causes. To further progress this investigation, we call for development in the following areas.

The first area relates to understanding the selective pressures on cooperative signals. The pioneering work of Sosis et al. (2007) showed that while collective ritual is omnipresent across the ethnographic record, the intensity and frequency of ritual performance co-vary with socio-ecological pressures. However, by relying on the Probability Sample File, their sample of cultural groups was necessarily limited. A phylogenetic analysis using well-calibrated trees for thousands of ethnolinguistic groups would provide a more robust test of the importance of various selective pressures (Šaffa et al., 2022). Likewise, combining new archaeological findings and progress in the genomic analysis might provide better-calibrated estimates of past hominin group sizes, which would pressure reliable cooperative communication.

Second, a more detailed understanding of neurocognitive mechanisms underlying the various ritual signals would help precisely estimate the possible presence of individual signals in the hominin evolutionary history. This is especially true for commitment signals and their often costly nature (Bulbulia, 2014), which rely on human psychological peculiarity (Sosis, 2003). Investigating the ontogeny of commitment, decisions about signaling modality, and signaler psychology should provide important insights into the functionality of commitment signals (Lang et al., 2022).

Finally, we aimed to keep as close as possible to the African archaeological record to provide a feasible transmission model for individual cooperative signals and their underlying neurocognitive mechanisms. This approach provides rather conservative estimates because earlier evidence for signals/mechanisms could usually be found outside of Africa, suggesting either convergent evolution of these traits, the dependency of expression of these traits on particular ecologies, or biased archaeological record. Nevertheless, limiting our analysis to the African continent still presents a relatively coarse-grained approach. It fails to consider the varied distribution of past hominins and their ecologies across the continent. While we wagered that collective ritual might have been a regular part of the early *H. sapiens*' social toolkit, the appearance and disappearance of the individual cooperative signals and their underlying neurocognitive mechanisms likely varied in African hominins pre-dating *H. sapiens*. Fine-grained data on partial hominin groups and their specific ecologies would make the proposed model of the evolution of collective ritual more precise. Together, these developments might lead to further exciting advances in understanding the most basic human social act.

Notes

1. Of course, such rituals have rich symbolic meaning nowadays (e.g., mythological rationale for each ritual sequence), and much scholarly attention has been dedicated to studying these meanings. However, we argue that underneath the veil of symbolic meanings, indexical relationships make ritual a functional communication platform.
2. Note that commitment signals do not necessarily function as commitment devices by foreclosing options to partake in a different action. While some commitment signals such as permanent markers (tattoos or scars) that are group specific preclude membership in other groups, commitment signals in our understanding refer to a broader set of signals that are transient and situationally specific.
3. We assume that cooperative intentions are not necessarily reflective of rigid types as specified in signaling games. Hence, intention to commit to a particular risky joint action may be graded and so is the signal cost.
4. Note that the theory of mind does not need to be employed through deliberate musings about others' minds through a symbolic communication code (although this is how contemporary humans primarily employ it).
5. We are grateful to Luke Glowacki for this insight.
6. Note that most of the used evidence is currently hotly debated (e.g., whether chimpanzee hunting is a cooperative action with role division) and for each trait that we tracked in the hominin lineage, the extant literature would often offer less conservative estimates (e.g., intentional fire use already in *H. ergaster*). Due to space constraints, we were unable to discuss each of these cases in detail but the selection of evidence used in this manuscript followed our general rule to include the most recent and well-established evidence.

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Data availability

The manuscript does not contain data.

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